The Arrow's Horizon: The Khronon's Cosmological Face — the Dark Fluid, the Clustering Dial, and One Cosmological Constant

Allen Proxmire

June 2026

Abstract

KM-I unified Event Density's gravity: one khronometric theory — Einstein at high accelerations, MOND below `a_0` — with the galactic phenomenology carried by the khronon, the substrate's arrow of time made dynamical. That left the cosmological half: the deep-vacuum soft spot, the standing MOND-class debts (clusters, the CMB), and a flagged two-faces question about Λ. This paper resolves the structure of all three. **First, the state space:** the configuration at which the deep-infrared sector degenerates, `(A, theta) = (0, 0)`, is exact Minkowski vacuum — **and ED's substrate forbids exact Minkowski vacuum: the tick never halts.** The degeneracy is diagnosed as an artifact of truncating the foliation EFT to the acceleration alone; the general sector `Wcal(A^2, Theta)` (`Theta = theta/3H_0`) is healthy at the physical cosmological point, and its `Theta`-dependence — the **regulator family** — is honestly unfixed. **Second, the orthogonality theorem:** static congruences have `theta == 0`, so every `Theta`-term vanishes identically in statics — *the entire tested sector of GR-I/KM-I (the modified Poisson equation, the lensing pass, the screening, the Cassini constraints) is invariant under the regulator choice.* The family is sequestered: no leak-back, no protection. **Third, the cosmological face itself:** on FRW the khronon's `Theta`-sector is a **dark fluid** `rho_Wcal(H)` with an automatic Λ-order late-time constant; and the condition for the dust-like clustering the CMB demands (`c_s ~= 0`) is **exactly the near-vacuum behavior previously flagged as the soft spot** — *the degeneracy cure and the clusters/CMB debt are the same dial*, which the regulator member must set while threading caustics, Čerenkov, BBN, and FRW stability. The class realizes both ends of the dial (mimetic dust at `c_s = 0`; integration-constant dust in the neighboring projectable branch), and the abundance-as-initial-condition character of those mechanisms is consonant with ED's value-inheritance. **Fourth, one Λ:** the khronon's vacuum term and the corpus's V1-backreaction Λ (Paper_038.5) carry the *same scale*, `H_0^2c^2/G`, by each paper's own statement — automatically, because both are projections of the one SCBU boundary `R_H = c/H_0` — and the level diagnosis (substrate mechanism vs EFT constant) converts double-counting into a category error: **one Λ, two descriptions**, with the matching computation named as the open derivation. The khronon's portfolio closes into one shape: **one field, one scale, five roles** — the arrow, the Einstein sector, the MOND sector, the dark-fluid dial, and the Λ-order vacuum term. Clusters and the CMB remain owed until the dial is set and tested; this paper reduces them from an inherited shortfall to one function under five filters, with the data as judge.

---

1. Introduction

1.1 What this paper does

KM-I ended with the galactic half of the unification derived and passing, and three cosmological items open: the deep-vacuum degeneracy (the one soft spot of the stability analysis), the inherited MOND-class debts (clusters, the CMB), and a flagged reconciliation between two Λ-mechanisms in the corpus. This paper supplies the structure of the cosmological half in four steps: the state-space resolution of the degeneracy (§3), the orthogonality theorem that sequesters the cosmological sector from everything already tested (§4), the FRW analysis — the dark fluid and the clustering dial (§5–§6), and the one-Λ thesis (§7).

Three orienting statements, up front:

- **No derivational hinge is exercised here.** KM-I's single hinge (the deep-IR matching to Paper_030) is inherited; this paper's results are a state-space observation, a kinematic theorem, a structural mapping, and a level identification. Where something is a family, it is labeled a family; where a computation is open, it is named.
- **The headline results:** (i) ED's substrate forbids the configuration at which the theory would degenerate (*no Minkowski vacuum — the tick never halts*); (ii) the cosmological sector is **provably invisible** to every test the companion papers passed (the orthogonality theorem); (iii) the dust-clustering condition the CMB demands coincides with the flagged soft spot — **the degeneracy cure and the clusters/CMB debt are one dial**; (iv) the corpus has **one Λ, not two** — the khronon's vacuum term is the EFT face of Paper_038.5's substrate-level boundary integral.
- **What the reader should *not* expect:** a chosen regulator member, a computed CMB spectrum, a derived Λ value, or a discharged cluster debt. The paper's product is the *exact specification* of the remaining model-building task: one function, five filters, data as judge.

1.2 Why this matters

Clusters and the CMB are where MOND-class theories go to die: galactic dynamics they handle, but the acoustic-peak structure wants something that clusters like cold dust at recombination, and the cluster masses want more than the modified force law supplies. Most relativistic MOND constructions face this with nothing to offer but epicycles. The khronon construction is differently positioned, for a structural reason this paper makes precise: **the same field already carries a cosmological sector** — not added for the purpose, but forced into existence by the EFT generality of the foliation sector and anchored, like everything else about the khronon, to the one scale `H_0`. Whether that sector *delivers* is a model-building and observational question; that it *exists, is sequestered from the tested physics, and sits exactly at the dust-clustering edge* is structure, and it is this paper's content.

1.3 How this fits the arc

- **Paper GR-I:** the bandwidth metric and lapse; the weak-field Einstein tests. *(inherited)*
- **Paper GR-II:** the khronometric class; the khronon; GW and Lorentz safety. *(inherited)*
- **Paper KM-I:** the galactic unification — the khronon's deep field as the MOND sector. *(inherited; this paper is its cosmological companion)*

- **Paper_038.5:** Λ as the V1 vacuum integral over `R_H = c/H_0`. *(read from source; unified with, not displaced)*
- **SCBU:** the one boundary `R_H = c/H_0`. *(here acquiring its second projection — and its dynamical carrier's vacuum term)*

1.4 Conventions and regime of validity

Signature `(-,+,+,+)`; `c = 1` where convenient; `Theta == theta/3H_0`, `A^2 == a_mua^mu/a_0^2`, `x == |grad Phi|/a_0` as in KM-I. The static results referenced are KM-I's (weak-field, quasi-spherical, DCGT limit); the cosmological statements here are **FRW-background structural results** — fluid content, scaling behaviors, and requirement maps, **not** computed spectra or fits. Exact coefficients are convention-dependent and carried schematically.

---

2. Primitive Inputs

Substrate primitives (postulated, Paper_087): P11/P13 (the arrow and the tick — the khronon's origin and, in §3, the no-Minkowski-vacuum observation), P04 (bandwidth — metric and lapse via GR-I), P02/P05 (adjacency/transport).

Inherited results (load-bearing): GR-I (metric, lapse, Einstein branch); GR-II (khronometric class, the khronon, cosmic-frame anchoring, `c_T = c`); KM-I (the foliation action with the acceleration sector `W(A^2)`, the deep-IR matching, the lensing/screening passes, the IR Cancellation Requirement); Paper_029 (`a_0 = cH_0/2pi`); Paper_038.5 (Λ as the V1 boundary integral; M3 tier; the value `rho_Lambda = (3/8pi)H_0^2M_P^2`).

External dictionary (inherited, labeled I): khronometric/aether FRW cosmology (the `lambdatheta^2` sector renormalizing `G_{\rm cosmo}`; BBN bound); **mimetic dark matter** (a unit-timelike-gradient scalar contributing exact pressureless dust, `c_s = 0`; caustics; the imperfect-fluid refinements); **dark matter as an integration constant** (projectable Hořava — the *neighboring* branch); the classical requirement that the CMB's acoustic structure be fit with a dust-like clustering component.

No new primitive is introduced; no primitive forcing is invoked. The paper exercises no derivational hinge of its own (§1.1).

2.5 Load-Bearing Step Audit

Status legend: **I** = inherited (cited paper or established external result); **D** = derived here; **D-structural** = derived from structure alone; **family** = honestly-unfixed functional freedom; **form-tier** = order-of-magnitude/structural consistency, no value claimed; **thesis** = a structurally-indicated identification with a named open computation; **open** = declared open.

Step	Status	Source / justification
Khronon, class, anchoring; galactic unification	I	GR-II; KM-I

Step	Status	Source / justification
Deep-IR sector degenerates at `(A,theta) = (0,0)`	I	KM-I §6/§7 (the flagged soft spot)
`(A,theta) = (0,0)` = exact Minkowski vacuum, **outside ED's state space**	**D-structural**	§3.1 — P11/P13: the tick never halts; the khronon's background is the Hubble flow
Pure-`A^2` form is a truncation; `Wcal(A^2,Theta)` is the general same-order sector	**D (EFT generality)**	§3.2
Generic `Wcal` healthy at the physical point `(0, Theta~1)`	**D (EFT generality)**	§3.2
The `Theta`-dependence of `Wcal`	**family**	§3.2 — the regulator family, labeled
Orthogonality theorem: `theta == 0` in statics $\Longrightarrow$ tested sector invariant under the regulator	**D**	§4 — kinematics; the paper's cleanest result
FRW: `lambdatheta^2 ~ lambdaH^2` renormalizes `G_{\rm cosmo}` (BBN-bounded)	I	§5 — class literature
FRW: `Wcal(0,Theta)` is a dark fluid `rho_Wcal(H)`	**D-structural**	§5
Late-time constant `~ a_0^2c^2/G ~ H_0^2c^2/G` automatic	**form-tier**	§5 — the single-scale argument (KM-I §7.4)
Requirements map (BBN / CMB-dust / clusters / late-Λ / Čerenkov+stability)	**D (mapping)**	§5
Dust-clustering condition `c_s ~= 0` = the flagged near-vacuum behavior	**D-structural**	§6 — **the soft spot is the doorway**; the central result
The three-way trade (clustering vs caustics vs Čerenkov); the window	**D (mapping)**	§6
Mimetic dust (`c_s = 0` end); integration-constant dust (projectable branch)	**I — directions**	§6 — proofs-of-concept; the latter is the *neighboring* branch
Abundance as initial condition $\leftrightarrow$ value-inheritance	**consonance noted**	§6
Both Λ faces carry `H_0^2c^2/G`	**documentary**	§7 — by each paper's own statement
Scale agreement automatic (shared SCBU anchor)	**D-structural**	§7 — two projections of one boundary

Step	Status	Source / justification
`Wcal_0` = EFT encoding of the V1 integral (**one** Λ)	**thesis**	§7 — matching computation named open; falsifiable edge stated
Matter-sector CC problem	**untouched**	§7 — stays where 038.5 left it
A regulator member chosen; a spectrum computed; clusters/CMB discharged	**NOT DONE / NOT CLAIMED**	preamble 2, 6

---

3. The State Space and the Regulator Family

3.1 ED has no Minkowski vacuum

KM-I's one soft spot was the deep-vacuum degeneracy: the acceleration sector's stiffness vanishes as `A -> 0`. The configuration at which the structure actually fails is `(A, theta) = (0, 0)` — zero acceleration *and* zero expansion: **exact Minkowski vacuum, eternal and inert. And ED's substrate forbids exact Minkowski vacuum: the tick never halts.** The khronon's background is the Hubble flow (GR-II); commitment does not cease and the tick does not stop (P11, P13). The physical background manifold is the `(A, theta)` half-space with `theta ~ H > 0`; the degenerate point sits on its excluded boundary. This is a *domain* observation — nothing about the action is modified — and it is one of the construction's strongest structural facts: *the substrate's refusal to ever stop ticking is what protects its gravity theory.*

3.2 The truncation diagnosis and the family

The remaining question is whether perturbations around the *physical* near-vacuum `(A -> 0, theta = 3H)` are healthy — and that exposes the deeper point: carrying the deep-IR sector as `W(A^2)` alone was a **truncation**. The foliation EFT's invariants at this order are the congruence scalars — acceleration, expansion, shear — and nothing singles out the acceleration as the sole argument of the non-analytic infrared structure. The general same-order sector is

$$W(A^2) \;\longrightarrow\; \mathcal{W}(A^2, \Theta), \qquad \Theta \equiv \frac{\theta}{3H_0},$$

and at the physical cosmological point `(A, Theta) = (0, \sim 1)` the generic member of this class sits at a **generic point of its domain**: there is no structural reason for its quadratic forms to vanish there, so the expansion sector supplies the stiffness the acceleration sector loses. The degeneracy is thereby diagnosed: **an artifact of the truncation, evaluated at a point ED never visits.**

The honest label, as always: the `Theta`-dependence is **not forced** — not by the substrate, not by the KM-I matching (which constrains only the static/`A` sector). It is the **regulator family**, and the next section shows why no member of it needs choosing for anything so far tested.

---

4. The Orthogonality Theorem

KM-I derived that static congruences have `theta = sigma = 0` identically. Therefore:

> **Every `Theta`-dependent term in `Wcal` vanishes identically in the static sector.** The entire tested body of the unification — the modified Poisson equation, the GR limit, the Combination-Rule and BTFR embedding, the lensing pass, the AQUAL ellipticity, the screening, and the Cassini constraints — is evaluated at `Theta`'s static value and is **invariant under the choice of regulator.**

Both directions of the consequence matter:

- **No leak-back.** The regulator cannot be tuned to assist the galactic physics — it is invisible there. Whatever member the cosmological data eventually select, KM-I's results stand unchanged. The retrofit worry is not resisted but *structurally impossible.*
- **No protection.** Equally, the tested sector cannot vouch for the regulator: the cosmological sector faces its own filters (§5–§6) on its own. Two independently falsifiable sectors — the healthiest structure available.

---

5. The Cosmological Face: the Dark Fluid

On FRW the aligned khronon is the comoving congruence (`a = 0`, `sigma = 0`, `theta = 3H`), and the foliation sector's cosmological roles are:

1. **`lambdatheta^2 ~ lambdaH^2`** — a renormalization of the effective cosmological Newton constant; the known khronometric effect, bounded by BBN's constraint on `G_{\rm cosmo}/G_N`. [I.] Inherited; nothing new.
2. **`Wcal(0, Theta)` — a dark fluid.** A function of `Theta = H/H_0` enters the Friedmann equation as a fluid `rho_Wcal(H)` with two automatic features: at late times (`H -> H_0`) it tends to a **constant of order `a_0^2c^2/G ~ H_0^2c^2/G`** — the Λ-order vacuum term (form-tier; §7) — and at early times its behavior is the regulator member's choice (a pure `Theta^2` piece folds into the `G_{\rm cosmo}` renormalization; other dependencies give other effective equations of state). **The family lives here.**
3. **The perturbation sector `deltaWcal`** — what the CMB and structure formation actually probe; §6.

The requirements map. For the sector to discharge the inherited debts, qualitatively:

Probe	Requirement on the `Theta`-sector
BBN	total `H^2`-tracking contribution inside the `G_{\rm cosmo}/G_N` bound
CMB peaks	a component of the right abundance clustering **like pressureless dust** (`w ~= 0`, `c_s ~= 0`) at recombination
Clusters	the same dust-like component supplying the mass MOND alone cannot
Late times	the Λ-order constant *(automatic)*

Probe	Requirement on the `Theta`-sector
Čerenkov / stability	propagating modes fast enough along cosmic-ray paths; FRW perturbations healthy

The hard row is the CMB's: it is where MOND-class theories historically die.

---

6. The Clustering Dial: the Soft Spot Is the Doorway

The paper's central structural result:

> **The condition for dust-like clustering is `c_s ~= 0` — and a near-zero sound speed at small `A` is exactly the near-vacuum behavior flagged as the deep-IR soft spot.** The feature that looked like the theory's vulnerability (the slowing, weakly-propagating near-vacuum scalar) is precisely what a dark-matter-like cosmological component *requires*: a fluid that does not free-stream, does not oscillate acoustically, and falls into potential wells like dust. **The degeneracy cure and the clusters/CMB debt are the same dial.**

The trade the regulator member must thread:

- **Too stiff** (`c_s ~ 1` near vacuum): degeneracy over-cured, Čerenkov trivially safe — but nothing clusters; the CMB debt stands in full.
- **Too soft** (`c_s = 0` exactly): maximal dust-like clustering — but caustics form (the known pathology of pressureless scalar dust) and strong coupling returns.
- **The window** (small but nonzero `c_s` in the cosmological regime): dust-like clustering on CMB/cluster scales with caustics regulated — the trade the *imperfect-fluid* refinements of the class navigate.

So "choosing the regulator member" is not bookkeeping: **it is the single choice that simultaneously sets the degeneracy cure, the Čerenkov compliance, and the theory's answer to clusters and the CMB.** One dial, three previously-separate ledger items.

The class realizes both ends of the dial [I — directions, not verifications]: **mimetic dark matter** (a constrained unit-gradient scalar — structurally a constrained khronon — contributing *exact* pressureless dust at `c_s = 0`, with the known caustic issues that motivated imperfect-fluid refinements), and **dark matter as an integration constant** (projectable Hořava's global-only Hamiltonian constraint leaving a dust component — with the honesty flag repeated: this is the *neighboring* branch of the family, a proof-of-concept that foliation gravity can manufacture dust, not automatically ED's mechanism).

A consonance, noted carefully. In both mechanisms the dust *abundance* is an initial condition or integration constant — not predicted. Under ED's own methodology this is not a defect but a classification: **the existence of the dust channel would be form-level structure; the observed abundance would be value-inherited** — the same form/value split the corpus applies to every constant. The class's "weakness" is ED's standing epistemic posture.

---

7. One Λ: the Boundary's Vacuum Imprint, at Two Levels

The corpus carried two Λ-mechanisms, flagged in KM-I §7.4 as faces to be reconciled. Read from source:

- **The V1 face (Paper_038.5; substrate level).** Λ as the V1 kernel's **UV-finite** vacuum integral (the form factor cuts off at `l_P`; no mode-tower divergence) over the cosmological boundary `R_H = c/H_0`; verdict M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED), the value landing at `rho_Lambda = (3/8pi)H_0^2M_P^2 ~ H_0^2c^2/G`. Its own honesty stands untouched: the naïve cutoff estimate failed by `~10^{60}` and is preserved as a record; the matter-sector hierarchy is open, reduced to Route-A closure.
- **The khronon face (KM-I §7.4 / §5 here; EFT level).** The foliation sector's vacuum term, `rho_Lambda ~ a_0^2c^2/G ~ H_0^2c^2/G`, by the single-scale argument. Form-tier; `Wcal_0` undetermined.

The scale identity is documentary — both faces carry `H_0^2c^2/G` by each paper's own statement — **and automatic**: both are projections of the *one SCBU boundary*, the **kernel projection** (V1's integral over `R_H`) and the **foliation projection** (the khronon's vacuum term at its `H_0` background). Agreement is what one boundary under two descriptions must produce; it is common origin, not coincidence, and not double-counting.

The level diagnosis and the thesis. The faces are not peers in one action: one is a *substrate-level mechanism*, the other an *EFT-level constant*. By standard effective-theory logic — low-energy constants *encode* the underlying theory rather than adding to it (the chiral-perturbation-theory pattern) —

> **One Λ, two descriptions:** the khronon's vacuum constant `Wcal_0` is the **EFT encoding of the substrate-level V1 backreaction integral.** Λ is the SCBU boundary's vacuum imprint — computed at the substrate by the V1 kernel (Paper_038.5), carried in the effective theory by the khronon (this arc).

Status, tiered: *structurally indicated, not proven.* The matching computation — deriving `Wcal_0` from the V1 integral — is the **named open derivation** of the cosmology thread, and the thesis has a clean **falsifiable edge**: if the substrate-side integral and the EFT-side constant (as the data constrain it through §5's fluid) ever disagree in sign or beyond `O(1)`, the framework has a genuine internal tension.

The clamps: the matter-sector cosmological-constant problem is untouched (the khronon face polices only the foliation sector's own term); no `O(1)` is derived on either face; value-inheritance and the `Theta_{ED}` think-don't-chase discipline stand.

---

8. Position — the Khronon's Portfolio

With this paper the khronon's roles close into one shape:

> **One field, one scale (`H_0`), five roles:** the **arrow of time** (GR-II — its origin), the **Einstein sector** at high accelerations (GR-I — enforced by commitment-linearity), the **MOND sector** below `a_0` (KM-I — the deep field), the **dark-fluid clustering dial** on FRW (§5–§6 — the clusters/CMB sector), and the **Λ-order**

vacuum term** (§7 — the boundary's imprint). Every role is anchored to the same number because the field has only one number to offer.

Relations: this paper **completes the cosmological half** KM-I opened (its §7.4 flag is resolved; its soft spot is re-classified as the clustering dial); it **unifies with, and does not displace, Paper_038.5** (one Λ, two levels — 038.5 remains the substrate mechanism and keeps its verdict tier); and it gives **SCBU** its second dynamical statement: the boundary `R_H = c/H_0` now has both a carrier field (KM-I) and a vacuum imprint computed at one level and carried at the other.

---

9. The Wedge — Empirical Content

1. **The CMB as the dial's judge.** The sharpest future test of the whole construction: the chosen regulator member must produce dust-like clustering of the right abundance at recombination, with the *same* function then fixed for clusters, Čerenkov, and BBN. A function with five filters has nowhere to hide.
2. **The galactic sector is already locked.** By the orthogonality theorem, nothing the cosmological data select can retroactively adjust the tested galactic physics — so any future CMB success cannot be a retrofit of the MOND sector, and any failure cannot be repaired from it. The two sectors test each other's honesty.
3. **The Λ-matching computation** — a derivation with a falsifiable outcome (sign / `O(1)` consistency between the substrate integral and the EFT constant).
4. **Standing wedges inherited:** wide binaries near `a_0` and the dynamical τ-sector (KM-I); the scalar GW polarization (GR-II).
5. **Not claimed as wedges:** a Λ value; a CMB fit; the dark-matter abundance (value-inherited if the channel exists).

---

10. Falsification Criteria

10.1 The window closes

Falsifier sentence: *A demonstration that no `Wcal(A^2, Theta)` member simultaneously satisfies the five filters — BBN, dust-like clustering at recombination, caustic regulation, Čerenkov, FRW stability — while remaining `Theta`-trivial in statics would falsify the §6 resolution: the clusters/CMB debt would return to the ledger as undischargeable within this construction.*

10.2 The CMB rejects every dial setting

Falsifier sentence: *CMB data requiring properties of the dust-like component that no `rho_Wcal(H)` fluid with near-zero sound speed can produce (given the galactic-side sector already fixed by KM-I) would falsify the khronon's cosmological face as the address for the MOND-class debts.*

10.3 The Λ faces disagree

Falsifier sentence: *Performance of the matching computation showing the substrate-level V1 integral and the EFT-level `Wcal_0` disagree in sign or beyond `O(1)` would falsify the one-Λ thesis (§7) and expose an internal tension between Paper_038.5 and this arc.*

10.4 BBN

Falsifier sentence: *A required regulator member whose `H^2`-tracking contribution violates the `G_{\rm cosmo}/G_N` BBN bound would falsify that member — and, if all clustering-capable members so violate, the construction's cosmological face.*

10.5 Sequestration breached

Falsifier sentence: *Any demonstration that a `Theta`-sector choice alters a static-sector observable (the modified Poisson equation, lensing, screening) would contradict the orthogonality theorem — a kinematic result — and would indicate the construction's congruence analysis is wrong at root.*

10.6 Standing fronts (inherited)

Falsifier sentence: *KM-I's falsification criteria (lensing slip; the Combination-Rule embedding; wide binaries; the interpolation family; `alpha_1, alpha_2`) carry over unchanged; this paper adds no protection against any of them.*

---

11. Position Statement

The khronon's cosmological face is structured, sequestered, and sits exactly where the data will judge it. The deep-vacuum degeneracy is resolved without retrofit — the degenerate point is outside ED's state space (no Minkowski vacuum: the tick never halts), and the pure-acceleration form was a truncation whose generic completion is healthy at the physical cosmological point. The completion's `Theta`-dependence is an honest family, **provably invisible to everything the companion papers tested** (the orthogonality theorem). On FRW that family is a dark fluid with an automatic Λ-order late-time constant; its perturbation sector sits — by the same near-vacuum behavior once flagged as the soft spot — at the dust-clustering edge the CMB demands: **the degeneracy cure and the clusters/CMB debt are one dial**, with the class realizing both ends and the abundance entering as a value-inherited initial condition. And the corpus has **one** Λ: the khronon's vacuum term is the EFT face of Paper_038.5's substrate-level V1 integral over the one SCBU boundary — the matching computation named, the falsifiable edge stated, the matter-sector problem left where it was.

What this paper claims. The structure: the state-space exclusion, the truncation diagnosis, the orthogonality theorem, the dark-fluid form with its requirements map, the dial identification, and the one-Λ thesis with its named computation.

What this paper does not claim. A chosen regulator member; a computed spectrum; a discharged cluster or CMB debt; a derived Λ value or abundance; a dust mechanism shown to live in ED's own branch; or any change to the matter-sector cosmological-constant problem.

Series context. Paper KM-II of the post-pivot curvature-emergence line — the cosmological companion to KM-I, closing the structural analysis of the unification. What remains is model-building (set the dial), one named derivation (the Λ matching), the guarded primitives question, and the sky.

KM-I said: one scalar, one scale, one gravity. KM-II completes it: **one field, one scale, five roles — and the data as judge.**

Appendix: Cross-References, Glossary, Reader Map

Cross-references

- **Paper_087:** position paper. — **Papers_028/029:** the boundary `R_H = c/H_0`; `a_0`.
- **Paper_038.5:** Λ as the V1 vacuum integral (the substrate face; unified with in §7).
- **Paper GR-I / GR-II / KM-I:** the metric and lapse; the class and the khronon; the galactic unification. *(inherited)*
- **SCBU:** the one boundary — two projections (§7), one carrier (KM-I).

Glossary

- **Regulator family.** The `Theta`-dependence of the general foliation sector `Wcal(A^2, Theta)`; unfixed, sequestered, five-filtered.
- **Orthogonality theorem.** `theta == 0` in statics $\Longrightarrow$ the regulator family is invisible to every tested static result. No leak-back; no protection.
- **Dark fluid `rho_Wcal(H)`.** The `Theta`-sector on FRW: a fluid whose density is a function of the expansion rate; Λ-order constant at late times.
- **The clustering dial.** The near-vacuum sound speed: too stiff = no clustering; too soft = caustics; the window = dust-like clustering with regulation. The same dial that cures the degeneracy.
- **The soft spot is the doorway.** The R3-flagged near-vacuum slow-mode behavior is the dust-clustering condition the CMB demands.
- **One-Λ thesis.** `Wcal_0` as the EFT encoding of Paper_038.5's substrate-level V1 boundary integral: one constant, two descriptions; matching computation open.
- **Mimetic dust / integration-constant dust.** Class proofs-of-concept at the dial's `c_s = 0` end and in the neighboring projectable branch, respectively. Directions, not verifications.
- **No Minkowski vacuum.** ED's substrate forbids `(A, theta) = (0, 0)`: the tick never halts. The degenerate point is excluded by the ontology.

Reader map and open work

Where to look next. - *The galactic half (the deep field, lensing, screening):* KM-I. — *The field and the class:* GR-II. — *The metric:* GR-I. - *The substrate face of Λ:* Paper_038.5. — *The boundary:* SCBU / Papers_028–029.

Open work (declared): the **Λ-matching computation** (`V1` integral → `Wcal_0`) — the cosmology thread's remaining derivation; **setting the dial** — the minimal `Wcal(Theta)` parametrization under the five filters (model-building); the **CMB/cluster confrontation** of the chosen member (observation); the guarded **primitives-origin** of the deep-IR branch; and KM-I's standing galactic fronts (wide binaries; the τ-sector; `alpha_1, alpha_2`).

End of Paper KM-II.

Post-pivot curvature-emergence line, the cosmological companion to KM-I. The deep-vacuum degeneracy is excluded by ED's own ontology (no Minkowski vacuum — the tick never halts) and diagnosed as a truncation artifact; the regulator family is sequestered by the orthogonality theorem (provably invisible to every tested result); the khronon's FRW face is a dark fluid with an automatic Λ-order constant whose perturbation sector sits at the dust-clustering edge — the soft spot is the doorway, and the degeneracy cure and the clusters/CMB debt are one dial under five filters. The corpus has one Λ: the khronon's vacuum term is the EFT encoding of the V1 boundary integral (matching computation named; falsifiable edge stated). One field, one scale, five roles — arrow, Einstein sector, MOND sector, clustering dial, vacuum term — and the data as judge. Clusters/CMB not discharged; no values derived; no member chosen.